

Simon Vutov

SimonVutov1@gmail.com | +1 613-606-7644 | linkedin.com/in/svutov | github.com/SimonVutov | simonvutov.github.io

EXPERIENCE

Software Engineer Intern - Deephaven

May 2026 - Aug 2026

TypeScript, React, Java, Python, Playwright, CI/CD

Remote

- Built a Java and React system for targeted UI banners with a persistent internal table, ACL enforcement, group-based resolution, CRUD event handling, dismissals, and timed expiration.
- Designed and shipped 2D splitter-intersection resizing for deeply nested Golden Layout panels, supporting three- and four-way crossings with nine automated drag tests.
- Cut cross-browser end-to-end CI runtime from **53 to 22 minutes** by sharding Playwright suites across six jobs and merging reports while preserving coverage.

Firmware Developer - UW Orbital - Satellite Design Team

Apr 2026 - Present

C, Python, ARM, CMake, GoogleTest, pytest

Waterloo, ON

- Built a lossless C/Python binary logging protocol for satellite firmware and ground-station tooling, reducing representative logs from **271 to 69 bytes (75%)** with byte-identical decoding.
- Added stream resynchronization, generated file-ID tables, runtime-selectable output, ARM cross-compilation, and **34 tests**; optimized alarm dequeue from $O(n)$ shifting to $O(1)$ pop.

Software Engineer Intern - BitGo

Sept 2025 - Dec 2025

Backend, Security, Prometheus, Grafana, Feature Flags

Palo Alto, CA (Remote)

- Shipped intent verification preventing blind signing in Wallet Platform threshold-signature transaction flows and designed reusable parsing across blockchain protocols.
- Instrumented verification failures with Prometheus and Grafana and deployed the safeguard behind a feature flag for safe staged rollout.

Software Engineer Intern - GenerativeModels.AI

Jan 2025 - Apr 2025

Python, FastAPI, React, Next.js, FAISS, Airflow

Toronto, ON

- Built FAISS-backed semantic search over **70k documents** with sub-800ms responses and Dockerized Airflow ingestion processing **900k product listings**.

PROJECTS

Invariant - C++ Numerical Computing Library | C++17, CMake, Unit Testing

2026

- Built a header-only template library with matrix/vector operations, Gaussian elimination, Jacobi iteration, polynomial interpolation, and stochastic-matrix analysis.

SimpleCar2 - Vehicle Physics Simulation | Unity, C#, Numerical Simulation

2025

- Engineered suspension, tire traction, torque curves, gear ratios, RPM, and angular acceleration; published a technical tutorial with **50k+ views**.

EDUCATION

Bachelor of Applied Science in Computer Engineering - University of Waterloo

Expected Apr 2029

Cumulative average: 86%; top 10% in ECE 250 Algorithms and Data Structures

Waterloo, ON

- Coursework: Algorithms and Data Structures, Computer Architecture, Digital Systems, Numerical Methods, Discrete Mathematics, Advanced Calculus.

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript/JavaScript, Java, Rust, C#, SQL

Systems and tools: Linux, Git, CMake, Docker, GDB, Valgrind, ARM, Playwright, CI/CD, Prometheus, Grafana

Frameworks and data: React, Next.js, FastAPI, REST APIs, PostgreSQL, FAISS, Airflow, PyTorch, CUDA